Achieving Envy-Freeness with Limited Subsidies under Negative Dichotomous Valuations

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Abstract

We study the problem of allocating indivisible *chores* among agents in a fair manner, with subsidies. Envy-free allocations of indivisible items are not guaranteed to exist, but envy-freeness can be achieved by additionally providing some subsidy to the agents; in the chores setting this subsidy is naturally read as compensation paid to the agents who absorb the workload. We study the subsidy required under *negative dichotomous* valuations, i.e., among agents whose marginal disutility for any chore is either zero or one: $v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\}$ for every agent i, every $S \subseteq M$ and every $g \in M \setminus S$. This is the exact mirror of the dichotomous goods model of Barman et al. [BKNS22], who prove that there a per-agent subsidy of 0 or 1 always suffices, and we conjecture that the same guarantee holds for chores.

The mirror is not a relabelling. Subsidies are one-signed — money is always a good — so negating an instance negates the valuations but not the money, and the goods theory does not transfer by symmetry. This report sets out what does transfer and what the chore problem looks like once it has. The Halpern–Shah characterisation of envy-freeable allocations [HS19] survives verbatim, since it assumes nothing about the sign or the structure of the valuations, and it delivers an *integral* minimum subsidy vector here; the matching lower bound of $n-1$ survives as well. Existing work already gives a bounded subsidy: negative dichotomous valuations are doubly monotone under the two-sided normalisation, so the reduction of Kawase et al. [KMS+24] yields $n-1$ per agent and $n(n-1)/2$ in total. The open question is whether the far stronger $\{0, 1\}$ guarantee of [BKNS22] survives the passage from goods to chores, a factor of n below that baseline.

We then isolate three structural facts about the chore problem: an arc-update lemma reducing the entire goods/chore asymmetry to two lines; a two-tier characterisation of the allocations admitting a $\{0, 1\}$ subsidy; and a size-shift transform exhibiting the chore problem as *exactly* the goods problem of [BKNS22] together with a cardinality-balancing requirement, which is the precise sense in which the conjecture does not follow from that work as a black box.

1 Introduction

Discrete fair division studies the allocation of resources that cannot be fractionally assigned among agents with individual preferences, and sits at the interface of mathematical economics and computer science [BCE+16, End17]. The classical fairness criterion is *envy-freeness* [Fol67, Var74]: no agent should prefer the bundle assigned to another agent over her own. For indivisible items an envy-free allocation need not exist, as the standard one-item two-agent instance shows, and a large body of work therefore studies relaxations such as envy-freeness up to one good (EF1) [Bud11, LMMS04].

A second and complementary line of work retains *exact* envy-freeness and buys off the residual envy with money. Each agent i receives, in addition to her bundle, a subsidy $p_i \geq 0$ supplied by a third party, and one asks for an allocation together with a subsidy vector under which every agent weakly prefers her own bundle-plus-subsidy to that of every other agent. The question is how much money this requires. Maskin [Mas87] answered it for unit-demand agents with $m = n$: under the scaling in which no agent values any single item above 1, a total subsidy of $n - 1$ suffices. Halpern and Shah [HS19] moved to the multi-demand setting with arbitrary m, gave the characterisation of *envy-freeable* allocations that the whole line now runs on, showed that a subsidy of $(n-1)m$ is necessary and sufficient in the worst case when the allocation is handed to us, and conjectured that $n - 1$ suffices when we may choose the allocation. Brustle et al. [BDN+20] proved that conjecture in the stronger per-agent form — one dollar to each agent suffices for additive valuations, with the allocation simultaneously EF1 and balanced — and showed that for general monotone valuations $2(n-1)$ per agent suffices, so that the total subsidy is $O(n^2)$ and, remarkably, *independent of the number of items*. Barman et al. [BKNS22] then improved the monotone bound by a factor of n on the dichotomous subclass: when every marginal $v_i(S \cup \{g\}) - v_i(S)$ lies in $\{0, 1\}$, there is an allocation whose accompanying subsidy is exactly 0 or 1 for each agent, hence at most $n - 1$ in total, computable in polynomial time in the value-oracle model. Their bound assumes neither additivity nor submodularity nor even subadditivity, and it is tight: with n agents and a single unit-valued good, $n - 1$ agents must each be paid 1.

Chores. Every result above is about *goods*. This paper asks the mirrored question, in which every item is a *chore*: an object that an agent would rather not hold, so that acquiring it weakly decreases her utility. Chores are not an exotic variant. Shift rotations, on-call duty, teaching and refereeing load, committee service, maintenance tasks and the liabilities attached to an estate are all allocation problems in which the items are burdens rather than prizes, and in all of them money is already the standard instrument of repair — as overtime pay, hardship allowance, or a course-release budget. If anything the subsidy model is more natural for chores than for goods, since compensating somebody for taking on unwanted work needs less justification than paying somebody for the privilege of not receiving a prize.

Concretely, we study *negative dichotomous* valuations: for every agent i, every $S \subseteq M$ and every $g \in M \setminus S$,

$$v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\}.$$

Equivalently, each marginal $v_i(S \cup \{g\}) - v_i(S)$ is 0 or -1: every item is a chore for every agent, and taking on one more chore costs an agent either nothing or exactly one unit. This is the exact mirror of the model of [BKNS22], and we impose no further structure — in particular the valuations need not be additive, submodular, or subadditive.

Binary marginals are the natural model whenever a task either lands on an agent as a fresh unit of burden or is absorbed by a commitment she has already made. An engineer already staffing an on-call window absorbs a second incident in that window at no extra cost; a referee already reading for a venue absorbs one more paper from the same batch; a driver already routed through a district absorbs an extra stop along the way. Whether a given chore is free depends in an arbitrary way on the rest of the agent's assignment, which is exactly why the model must tolerate non-additive and even non-subadditive valuations, and it is the counterpart of the scheduling example used by [BKNS22] to motivate dichotomous valuations for goods.

Why this is not a change of sign. It is tempting to expect that a chores result follows from the corresponding goods result by negating the valuations. It does not, for three reasons that we record here because they shape everything that follows.

First, money is one-signed. The subsidy vector is constrained to $p \in \mathbb{R}^n_+$, and negating an instance negates the valuations but leaves the subsidies where they are. The polyhedron of feasible subsidy vectors is therefore a genuinely different object in the two models, and no formal duality carries a bound across.

Second, the obvious transformation is available only for two agents. Writing $c_i := -v_i$ for the cost form of a negative dichotomous valuation, the complement $\tilde{v}_i(S) := c_i(M) - c_i(M \setminus S)$ is again dichotomous in the sense of [BKNS22], so the class is closed under complementation. But the complement transformation respects the allocation only when $M \setminus A_j = A_i$, that is, only when $n = 2$; see Remark 9. For $n \geq 3$ there is no such reduction, and that is where the content of the problem lies.

Third, and most tellingly, envy flows the other way. Under goods, envy concentrates *into* the agent holding the valuable items: many agents envy one, and the tight instance of [BKNS22] pays $n - 1$ agents one dollar each so that they stop envying the single agent who received the good. Under chores, envy emanates *out of* the agent carrying the load: one agent envies many. Since the minimum subsidy to an agent is the weight of the heaviest directed path leaving her in the envy graph [HS19, Theorem 2], the two situations are not interchangeable, and one should not assume in advance that the same bound is the right target.

What is already known, and what is open. Two facts delimit the problem before any work is done, and we state them explicitly so that the contribution can be measured against them.

On the one hand, existence of a bounded subsidy is not in question. Negative dichotomous valuations are doubly monotone — every item is a chore for every agent, which is a special case of every item being a good or a chore for each agent — and they satisfy the normalisation $|v_i(S \cup \{e\}) - v_i(S)| \leq 1$ that [KMS+24, §2] imposes. The reduction of Kawase et al., which converts any EF1 allocation into an envy-free one, therefore applies off the shelf and yields a subsidy of at most $n - 1$ per agent and $n(n-1)/2$ in total [KMS+24, Theorem 1]. Two details make the containment work and are easy to get wrong. Their normalisation is two-sided, so it bounds chore disutility as well as value; and the EF1 they require is the two-sided form of Theorem 5, not the goods-only form of [HS19] — the two differ precisely when chores are present. An EF1 allocation is guaranteed to exist and to be computable in polynomial time for doubly monotone valuations, so the reduction has an input; for that step [KMS+24, §2] cites [LMMS04] and [BSV21], and the citation matters, because the envy-cycle argument of [LMMS04] does *not* port to chores unaltered and it is [BSV21] that supplies the repair.

On the other hand, the lower bound of the goods model survives intact.

Example 1 (The $n-1$ lower bound transfers)**.** Take n agents and $m = n-1$ chores, with $c_i(S) = |S|$ for every agent i — identical, binary, additive costs, which are in particular negative dichotomous. Every complete allocation leaves at least one agent empty-handed. Under the balanced allocation, in which $n-1$ agents hold one chore each and one agent holds nothing, each of the $n-1$ loaded agents envies the empty agent by exactly 1 and no other envy is present, so the minimum subsidy assigns 1 to each loaded agent and 0 to the empty one, for a total of $n-1$. No allocation does better: any complete allocation gives some agent a bundle of size $k \geq 1$ while some other agent is empty, and that agent's heaviest outgoing path already has weight k.

So the target statement is precisely the analogue of [BKNS22, Theorem 4].

Conjecture 2. *For every instance with negative dichotomous valuations there is an envy-free solution $(\mathcal{A}, p)$ with $p \in \{0,1\}^n$, hence with total subsidy at most $n-1$; and it can be computed in polynomial time given value-oracle access.*

Example 1 shows Conjecture 2 would be tight. Between the $n-1$ per agent that [KMS+24] already gives and the $\{0,1\}$ per agent being asked for lies a factor of n in the total, exactly the gap that [BKNS22] closed on the goods side. Our objective is to close it on the chores side or to exhibit an instance showing that it cannot be closed.

Scope of this report. Conjecture 2 is open. This report sets up the problem and stops there: Section 2 fixes the model and notation and records what transfers from the goods theory unchanged, and Section 3 isolates three structural facts about the chore problem — an arc-update lemma (Lemma 14) reducing the entire goods/chore asymmetry to two lines, a two-tier characterisation of $\{0,1\}$ solutions (Proposition 15), and a size-shift transform (Theorem 16) exhibiting the chore problem as exactly the goods problem of [BKNS22] together with a cardinality-balancing requirement. The last of these is the precise sense in which Conjecture 2 does not follow from [BKNS22] as a black box.

Work towards the conjecture is in progress and is deliberately not reported here. Several special cases are settled and two natural proof strategies have been ruled out by explicit machine-checked instances; none of it is stated below until it is worth stating.

Related Work. The subsidy line for goods is described above. Within it, the dichotomous subclass has attracted separate attention: Halpern and Shah [HS19] settle binary additive valuations with a total subsidy of $n-1$, Goko et al. [GIK+21] obtain the analogous bound for binary submodular valuations together with a strategy-proof mechanism, and Barman et al. [BKNS22] subsume both by dropping every structural assumption beyond binary marginals. Dichotomous preferences are themselves a well-studied restriction in social choice [BMS05, BEF21]. On the computational side, Caragiannis and Ioannidis [CI21] show that minimising the subsidy is NP-hard and give a fully polynomial-time approximation scheme for a constant number of agents, and Narayan et al. [NSV21] study the welfare cost of achieving envy-freeness with transfers.

The paper closest to the present setting is that of Kawase et al. [KMS+24], the only work in the envy-freeness-with-subsidy line that admits chores at all. Working with doubly monotone valuations under the two-sided normalisation, it decouples the two jobs that [BDN+20] performed together: rather than constructing a specific allocation and showing that it is cheap to subsidise, it takes *any* EF1 allocation as input and bounds the subsidy needed to convert it, obtaining $n-1$ per agent and $n(n-1)/2$ in total. Since EF1 allocations are known to exist for doubly monotone valuations, the wider class comes along at no cost. That reduction is what makes the present question well-posed rather than open-ended, and it is the baseline we are trying to beat.

The complementary repair — no money, weaker fairness notion — is pursued for binary valuations by Bu et al. [BSY23], who show that EFX allocations always exist and can be computed in polynomial time for general binary valuations, thus extending the matroid-rank result of Babaioff et al. [BEF21]. Under binary valuations one therefore has a clean dichotomy on the goods side: either pay 0 or 1 per agent and obtain exact envy-freeness [BKNS22], or pay nothing and obtain EFX [BSY23]. Whether either branch survives the passage to chores is open; this paper addresses the first.

The chores side. Work on indivisible chores has largely developed alongside rather than inside the subsidy line, and the two have met less often than one might expect. Where money has been brought to chores, it has been in the service of *proportionality* rather than envy-freeness: the question studied is the total subsidy needed to bring every agent down to her proportional share of the burden, for which the best bounds under additive disutilities normalised to at most 1 per chore are $n/4$ and then $n/3 - 1/6$, recently obtained together with Pareto optimality by Garg et al. [GSW25]. That is a different fairness notion and a different accounting; it does not bound the subsidy needed for envy-freeness.

The no-money branch, by contrast, has a direct chore analogue of [BSY23]. Tao et al. [TWYZ23] study chores with binary marginals and show that for binary *additive* costs an allocation that is simultaneously EFX and Pareto optimal always exists and is computable in polynomial time. So on the chores side the second half of the binary dichotomy — pay nothing, obtain EFX — is established at least for additive costs, while the first half, pay 0 or 1 per agent and obtain exact envy-freeness, is what Conjecture 2 asks for. That money is genuinely needed here is not obvious a priori but is known: Bhaskar et al. [BSV21] show that deciding whether an exactly envy-free allocation of chores exists is NP-complete already for binary additive costs, and give a polynomial-time EF1 algorithm for chores and for doubly monotone instances.

Where the two lines have met is on the *additive* side, and recently. Lu et al. [LMS26] prove the goods-and-chores analogue of [BDN+20]: for additive utilities in which each item is a good for some agents and a chore for others, with every $|v_i(g)| \le 1$, a subsidy of one dollar per agent suffices, the bound is tight, and the allocation is computable in polynomial time — hence $n-1$ in total. This settles the additive chore problem completely.

It does not settle ours, and the reason is exactly the reason [BKNS22] was not a corollary of [BDN+20]. Dichotomous valuations are not additive and additive valuations are not dichotomous; the two classes are incomparable. The four corners are

	goods	chores
additive	[BDN+20]	[LMS26]
dichotomous	[BKNS22]	**Conjecture 2**

each of the three settled corners giving one dollar per agent and $n-1$ in total. The present question is the fourth. Note also that [LMS26] does not displace [KMS+24] as the baseline of Section 1: for valuations that are dichotomous but not additive, $n-1$ per agent from [KMS+24] remains the best published bound.

We are not aware of any treatment of envy-freeness with subsidy for chores under dichotomous costs. This is the outcome of a literature search rather than a proof of novelty — [LMS26] appeared in July 2026, which is a reminder of how quickly the surrounding corners are being filled in — and it should be repeated before the paper is circulated.

2 Notation and Preliminaries

We study the allocation of m indivisible items among n agents, with subsidies. Write $N = [n]$ for the set of agents and M for the set of items, $|M| = m$. Each agent $i \in N$ has a valuation $v_i : 2^M \to \mathbb{R}$ with $v_i(\emptyset) = 0$, and we represent an instance by the tuple $\langle N, M, \{v_i\}_{i\in N}\rangle$. Algorithms operate in the standard *value-oracle* model: for each v_i we assume only an oracle that returns $v_i(S)$ for a queried set $S \subseteq M$, and running time is measured in n, m and the number of oracle calls.

An *allocation* $\mathcal{A} = (A_1, \dots, A_n)$ is an ordered tuple of pairwise disjoint *bundles*, with A_i assigned to agent i. The allocation is *complete* if $\bigcup_{i\in N} A_i = M$ and *partial* otherwise. The ordering matters throughout: much of the theory below concerns permuting a fixed family of bundles among the agents, and for a permutation $\sigma \in \mathbb{S}_n$ we write $\mathcal{A}_\sigma := (A_{\sigma(1)}, \dots, A_{\sigma(n)})$ for the allocation in which agent i receives $A_{\sigma(i)}$. The utilitarian welfare of $\mathcal{A}$ is $\mathrm{SW}(\mathcal{A}) = \sum_{i\in N} v_i(A_i)$.

Definition 3 (Envy-Freeness)**.** *An allocation $\mathcal{A} = (A_1, \dots, A_n)$ is* envy-free *(EF) iff $v_i(A_i) \geq v_i(A_j)$ for all agents $i, j \in N$.*

Envy-free allocations of indivisible items need not exist, so we allow a third party to supplement the allocation with money. Agents have quasi-linear utilities: agent i receiving bundle A_i and subsidy p_i enjoys utility $v_i(A_i) + p_i$, with money entering linearly and at the same exchange rate for every agent.

Definition 4 (Envy-Free Solution)**.** *An allocation $\mathcal{A} = (A_1, \dots, A_n)$ and a subsidy vector $p = (p_1, \dots, p_n) \in \mathbb{R}^n_+$ constitute an* envy-free solution *$(\mathcal{A}, p)$ iff $v_i(A_i) + p_i \geq v_i(A_j) + p_j$ for all $i, j \in N$.*

An allocation $\mathcal{A}$ is *envy-freeable* if some $p \in \mathbb{R}^n_+$ makes $(\mathcal{A}, p)$ an envy-free solution; this is a property of the allocation alone. Following [BKNS22] we write $\mathrm{M}(p) := \{i \in N \mid p_i \geq p_j \text{ for all } j \in N\}$ for the set of maximally subsidised agents.

We will occasionally need the standard relaxation of envy-freeness. Note that two inequivalent forms of it appear in this literature: the goods-only form of [HS19], which removes one item from the envied bundle, and the two-sided form used by [KMS+24], which removes one item from *either* bundle. They coincide when all items are goods and differ once chores are present, and only the two-sided form supports the doubly monotone results. We adopt the two-sided form.

Definition 5 (EF1, two-sided form)**.** *An allocation $\mathcal{A}$ is* envy-free up to one item *(EF1) iff for all $i, j \in N$ there exists $X \subseteq A_i \cup A_j$ with $|X| \leq 1$ such that $v_i(A_i \setminus X) \geq v_i(A_j \setminus X)$.*

2.1 Negative Dichotomous Valuations

Definition 6 (Negative Dichotomous Valuation)**.** *A valuation $v_i : 2^M \to \mathbb{R}$ with $v_i(\emptyset) = 0$ is* negative dichotomous *iff*

$$v_i(S) - v_i(S \cup \{g\}) \in \{0,1\} \qquad \text{for every } S \subseteq M \text{ and every } g \in M \setminus S,$$

equivalently iff every marginal $\Delta_i^+(S,g) := v_i(S \cup \{g\}) - v_i(S)$ lies in $\{-1,0\}$.

Every item is thus a chore for every agent: v_i is non-increasing, so $v_i(S) \geq v_i(T)$ whenever $S \subseteq T$. We stress that nothing further is assumed — the valuations need not be additive, submodular, or subadditive — which is the generality at which [BKNS22] works on the goods side.

It is often more convenient to work with the non-negative *cost form*

$$c_i := -v_i, \qquad \text{so} \qquad c_i(\emptyset) = 0, \quad c_i(S) \leq c_i(T) \text{ for } S \subseteq T, \quad c_i(S \cup \{g\}) - c_i(S) \in \{0,1\}.$$

Definition 7 (Dichotomous cost function)**.** *A set function $c : 2^M \to \mathbb{R}$ is* dichotomous *if $c(\emptyset) = 0$ and every marginal $c(S \cup \{g\}) - c(S)$ lies in $\{0,1\}$.*

Monotonicity is not a separate hypothesis: marginals in $\{0,1\}$ are non-negative, so a dichotomous c is automatically non-decreasing. We nonetheless list it above because it is the property most often used directly.

Remark 8 (Three adjectives that are routinely confused)**.** *Monotone* means only $c(S) \leq c(T)$ for $S \subseteq T$, with no bound whatever on how large an increment can be; it is far weaker than Theorem 7 and, on its own, supports almost none of the arguments below. *Dichotomous* adds the binary-marginal condition and is the class of this paper. *Binary additive* is the strictly smaller class $c_i(S) = |S \cap D_i|$ for a set D_i of disliked chores, i.e. additive with per-item costs in $\{0,1\}$; it is the case settled by [LMS26].

A warning about the literature: several papers say "binary valuations" for what is called dichotomous here — [BSY23] and [TWYZ23] both do — and reserve "binary additive" for the additive subclass. When a bound is quoted, check which is meant, since the classes are not the same and the gap between them is where this paper lives.

The correspondence $v_i \leftrightarrow c_i$ is a bijection between negative dichotomous valuations and dichotomous cost functions in the sense of [BKNS22]: the class we study is exactly the pointwise negation of the class they study. In cost form an envy-free solution is the requirement $c_i(A_i) - p_i \leq c_i(A_j) - p_j$ for all i, j, read as "my own burden net of my compensation is no worse than what I perceive of yours". We use whichever of v_i and c_i is clearer locally and always state which.

The negation, however, does not carry results across, for the reasons given in Section 1. The following remark isolates the one case in which a transformation does exist, and shows why it is uninformative.

Remark 9 (Complementation, and why it stops at $n = 2$)**.** For a negative dichotomous v_i with cost form c_i, define the *complement* $\tilde{v}_i(S) := c_i(M) - c_i(M \setminus S)$. Then $\tilde{v}_i(\emptyset) = 0$ and, for $g \notin S$, writing $T := M \setminus (S \cup \{g\})$,

$$\tilde{v}_i(S \cup \{g\}) - \tilde{v}_i(S) = c_i(M \setminus S) - c_i(T) = c_i(T \cup \{g\}) - c_i(T) \in \{0,1\},$$

so $\tilde{v}_i$ is dichotomous in the sense of [BKNS22]. The class is therefore closed under complementation, without any submodularity assumption. Now let $n = 2$ and let $\mathcal{A} = (A_1, A_2)$ be complete, so that $M \setminus A_2 = A_1$ and $M \setminus A_1 = A_2$. Then $-c_i(A_i) = \tilde{v}_i(A_j) - c_i(M)$ and $-c_i(A_j) = \tilde{v}_i(A_i) - c_i(M)$, so $(\mathcal{A}, p)$ is an envy-free solution for the chores instance $\{v_i\}$ if and only if $((A_2, A_1), p)$ is an envy-free solution for the goods instance $\{\tilde{v}_i\}$: swap the bundles and leave each subsidy attached to its agent. Two-agent chore division is thus exactly two-agent goods division, and [BKNS22] settles it. The step $M \setminus A_j = A_i$ has no analogue for $n \geq 3$, where the complement of one bundle is a union of several others, and no reduction of this kind is available.

2.2 The Envy Graph and the Halpern–Shah Characterisation

For an allocation $\mathcal{A}$, the *envy graph* $G_{\mathcal{A}}$ is the complete weighted directed graph on vertex set N in which the arc (i, j) carries weight

$$w_{\mathcal{A}}(i, j) \;:=\; v_i(A_j) - v_i(A_i) \;=\; c_i(A_i) - c_i(A_j),$$

the envy that agent i has towards agent j. The weight of a directed path P is $w_{\mathcal{A}}(P) = \sum_{(i,j)\in P} w_{\mathcal{A}}(i, j)$, and $\ell_{\mathcal{A}}(i)$ denotes the maximum weight of any path in $G_{\mathcal{A}}$ starting at i, the empty path of weight 0 included.

The two results below are the engine of the entire subsidy line. Both are stated by Halpern and Shah [HS19] for additive valuations, but their proofs use only the arc weights and permutations of the bundles — neither additivity nor monotonicity nor any sign condition on v_i enters — and [BKNS22] invokes them in exactly this generality for (non-additive) dichotomous valuations. They therefore apply verbatim to negative dichotomous valuations, and we use them without further comment.

Theorem 10 ([HS19])**.** *For any allocation $\mathcal{A} = (A_1, \ldots, A_n)$, the following are equivalent.*

(i) $\mathcal{A}$ is envy-freeable.

(ii) $\mathcal{A}$ maximises utilitarian welfare across all reassignments of its own bundles among the agents: for every permutation σ over N, $\sum_{i\in N} v_i(A_i) \geq \sum_{i\in N} v_i(A_{\sigma(i)})$.

(iii) The envy graph $G_{\mathcal{A}}$ has no positive-weight directed cycle.

Theorem 11 ([HS19])**.** *For any envy-freeable allocation $\mathcal{A}$, the subsidy vector p^* given by $p^*_i := \ell_{\mathcal{A}}(i)$ realises an envy-free solution $(\mathcal{A}, p^*)$, and $p^*_i \leq p_i$ for every $i \in N$ and every p such that $(\mathcal{A}, p)$ is an envy-free solution. It can be computed in strongly polynomial time by running an all-pairs longest-path computation on $G_{\mathcal{A}}$.*

Condition (ii) of Theorem 10 has the standing consequence that an envy-freeable allocation can always be manufactured from an arbitrary one: fix the bundles and reassign them according to a maximum-weight perfect matching between agents and bundles. What is at issue is never the existence of an envy-freeable allocation, only the size of the subsidy that Theorem 11 then charges for it.

Specialising to our model gives the following, which we use throughout.

Observation 12 (Integrality)**.** *Let $\{v_i\}_{i\in N}$ be negative dichotomous. Then $v_i(S) \in \mathbb{Z}_{\le 0}$ for every i and every $S \subseteq M$; every arc weight $w_{\mathcal{A}}(i,j)$ is an integer; and for every envy-freeable allocation $\mathcal{A}$ the minimum subsidy vector satisfies $p^* \in \mathbb{Z}^n_{\ge 0}$.*

Proof Fix i and $S = \{g_1, \ldots, g_k\}$. Telescoping from $v_i(\emptyset) = 0$ along any ordering of S writes $v_i(S)$ as a sum of k marginals, each of which lies in $\{-1, 0\}$ by Definition 6; hence $v_i(S) \in \mathbb{Z}$ and $-k \le v_i(S) \le 0$. Consequently each arc weight $w_{\mathcal{A}}(i,j) = v_i(A_j) - v_i(A_i)$ is a difference of integers, and the weight of any directed path, being a finite sum of arc weights, is an integer. By Theorem 11, $p_i^* = \ell_{\mathcal{A}}(i)$ is the maximum such weight over paths starting at i, and it is non-negative because the empty path has weight 0. □

Observation 12 is what makes the target statement — a subsidy vector in $\{0,1\}^n$ — a statement about a *bound* rather than about rounding: once p^* is known to be integral, showing $p_i^* \le 1$ for every i is the same as showing $p^* \in \{0,1\}^n$. The same observation underlies the corresponding step in [BKNS22], where the arc weights of a dichotomous instance are likewise integers.

A second normalisation costs nothing and removes a clause from the target.

Proposition 13 (Some agent is always unpaid)**.** *For every envy-freeable allocation $\mathcal{A}$ there is an agent h with $p_h^* = 0$. Consequently, if $p^* \in \{0,1\}^n$ then $\sum_{i \in N} p_i^* \le n - 1$.*

Proof By Theorem 10(iii) the envy graph $G_{\mathcal{A}}$ has no positive-weight cycle, so a walk of maximum weight may be taken simple and $\ell_{\mathcal{A}}$ is finite; it is non-negative because the empty path qualifies. Choose a path P attaining $\max_i \ell_{\mathcal{A}}(i)$ and let h be its final vertex. If $\ell_{\mathcal{A}}(h) > 0$ there is a positive-weight walk Q leaving h, and P followed by Q is a walk from P's start of weight $w_{\mathcal{A}}(P) + w_{\mathcal{A}}(Q) > w_{\mathcal{A}}(P)$, contradicting the maximality of P among walks. Hence $\ell_{\mathcal{A}}(h) = 0$. The second claim is immediate: at most $n-1$ coordinates of p^* can equal 1. □

Proposition 13 justifies the word "hence" in Conjecture 2: the bound on the *total* is not an independent requirement but a consequence of the per-agent one. The conjecture is therefore a single-clause statement,

$$\exists \mathcal{A} : \quad \max_{i \in N} \ell_{\mathcal{A}}(i) \le 1,$$

and it is this form that the rest of the report works with. Two cautions are worth recording alongside it. First, the bound cannot be improved: the family of Example 1 attains $n-1$ for every n, verified exhaustively for $n \le 7$. Second, the conjecture is *not* the assertion that the utilitarian-optimal subsidy is at most $n-1$. Minimising the total over all allocations can do strictly better than any allocation with $p^* \in \{0,1\}^n$, by concentrating the subsidy on a single agent: there are instances at $n = 4$ whose cheapest envy-free solution costs 2, borne entirely by one agent, while every allocation with $p^* \in \{0,1\}^4$ costs 3. The $\{0,1\}$ shape is a genuine restriction, not a normalisation of the total.

3 Structure of the Chore Problem

Throughout this section we work in cost form, so $w_{\mathcal{A}}(i,j) = c_i(A_i) - c_i(A_j)$.

3.1 The arc-update table, and the whole of the asymmetry

Every incremental algorithm in this literature adds one item to one bundle and then asks what happened to the envy graph. The answer is two lines long, and comparing those two lines against their goods counterparts accounts for the entire difference between the present problem and that of [BKNS22].

For an allocation $\mathcal{A}$, an agent x and an unallocated chore $g \notin \bigcup_i A_i$, write

$$Z^x \;:=\; (A_1, \ldots, A_x \cup \{g\}, \ldots, A_n)$$

for the allocation obtained by giving g to x, and put

$$\delta_x \;:=\; c_x(A_x \cup \{g\}) - c_x(A_x), \qquad \delta_{i,x} \;:=\; c_i(A_x \cup \{g\}) - c_i(A_x),$$

both of which lie in $\{0, 1\}$. So δ_x is what the chore costs x herself, and $\delta_{i,x}$ is what it costs in i's opinion.

Lemma 14 (Arc update). *For all $i, j \in N$,*

$$w_{Z^x}(i,j) = w_{\mathcal{A}}(i,j) \quad (i,j \neq x), \qquad w_{Z^x}(x,j) = w_{\mathcal{A}}(x,j) + \delta_x, \qquad w_{Z^x}(i,x) = w_{\mathcal{A}}(i,x) - \delta_{i,x}.$$

Proof Only A_x changes, so an arc between two agents other than x has both endpoints untouched. For an arc leaving x, $w_{Z^x}(x,j) = c_x(A_x \cup \{g\}) - c_x(A_j) = w_{\mathcal{A}}(x,j) + \delta_x$. For an arc entering x, $w_{Z^x}(i,x) = c_i(A_i) - c_i(A_x \cup \{g\}) = w_{\mathcal{A}}(i,x) - \delta_{i,x}$. □

The mirror statement for goods is obtained by replacing costs with values: there the arcs *out of* the receiving agent fall and the arcs *into* her rise. The consequences are opposite in a way that matters.

	goods [BKNS22]	chores (here)
arcs out of the receiver	fall	**rise by** δ_x
arcs into the receiver	**rise**	fall by $\delta_{i,x}$
so the receiver must be	*maximally* subsidised	*minimally* subsidised
which is guaranteed by	$\mathrm{M}(p) \neq \emptyset$	$p_x = 0$ for some x

Both guarantees hold trivially — some agent is always maximally subsidised, and some agent always has $p_x = 0$ under the minimum subsidy vector. The asymmetry is subtler than the table suggests. Under goods, the harm done by an insertion is confined to paths *ending* at the receiver, and requiring $x \in \mathrm{M}(p)$ caps every such path at $p_u - p_x \leq 0$ in one stroke. Under chores, the harm is done to paths *leaving* the receiver, and $p_x = 0$ caps only the paths that *start* at x. A path that merely passes through x — entering along an arc from some i with $\delta_{i,x} = 0$ and leaving along an arc that has just risen by $\delta_x = 1$ — is not controlled by any condition on p_x. Whether that gap can actually be realised is the question an incremental algorithm has to answer.

3.2 The shape of a $\{0,1\}$ solution

Proposition 15 (Two-tier characterisation)**.** *Let $p \in \{0,1\}^n$ and put $S := \{i \in N \mid p_i = 1\}$. Then $(\mathcal{A}, p)$ is an envy-free solution if and only if*

(a) $c_i(A_i) \le c_i(A_j)$ *whenever i, j lie both in S or both in $N \setminus S$;*

(b) $c_i(A_i) \le c_i(A_j) + 1$ *for $i \in S$ and $j \in N \setminus S$;*

(c) $c_i(A_i) + 1 \le c_i(A_j)$ *for $i \in N \setminus S$ and $j \in S$.*

Proof In cost form Definition 4 reads $c_i(A_i) \le c_i(A_j) + p_i - p_j$ for all i, j. Substituting the three possible values $0, 1, -1$ of $p_i - p_j$ gives (a), (b) and (c) respectively. □

The reading is worth keeping in mind when designing an algorithm. The unpaid agents are exactly envy-free among themselves and *strictly* prefer their own bundle to that of every paid agent, by a full unit; the paid agents are exactly envy-free among themselves and envy-free up to one unit towards the unpaid. This is a certificate format, and it also suggests choosing the partition $S \mid N \setminus S$ *first* and allocating afterwards, rather than building the allocation up one chore at a time.

3.3 The size-shift transform

Remark 9 showed that complementation relates the chore and goods models only for $n = 2$. There is a second transform, available for every n, which does not solve the problem but says precisely how far it is from being a corollary of [BKNS22].

Theorem 16 (Size shift)**.** *For a negative dichotomous instance with cost functions c_i, define $\tilde{v}_i(S) := |S| - c_i(S)$. Then each $\tilde{v}_i$ is a dichotomous* goods *valuation in the sense of [BKNS22], and for every allocation $\mathcal{A}$ and all i, j,*

$$w_{\mathcal{A}}(i,j) \;=\; \tilde{w}_{\mathcal{A}}(i,j) \;+\; |A_i| - |A_j|,$$

where $\tilde{w}_{\mathcal{A}}$ denotes the envy graph of the goods instance $\{\tilde{v}_i\}$. Consequently every cycle *has the same weight in both graphs — so $\mathcal{A}$ is envy-freeable for the chores $\{c_i\}$ if and only if it is envy-freeable for the goods $\{\tilde{v}_i\}$ — while for a path P from u to v,*

$$w_{\mathcal{A}}(P) \;=\; \tilde{w}_{\mathcal{A}}(P) + |A_u| - |A_v|, \qquad \textit{hence} \qquad \ell_{\mathcal{A}}(u) \;=\; \max_{v \in N} \Big[\, \tilde{d}_{\mathcal{A}}(u,v) + |A_u| - |A_v| \,\Big],$$

with $\tilde{d}_{\mathcal{A}}(u,v)$ the maximum weight of a path from u to v in the goods graph and $\tilde{d}_{\mathcal{A}}(u,u) = 0$.

Proof $\tilde{v}_i(\emptyset) = 0$, and for $g \notin S$, $\tilde{v}_i(S \cup \{g\}) - \tilde{v}_i(S) = 1 - (c_i(S \cup \{g\}) - c_i(S)) \in \{0,1\}$, so $\tilde{v}_i$ is dichotomous. For the identity,

$$\tilde{w}_{\mathcal{A}}(i,j) = \tilde{v}_i(A_j) - \tilde{v}_i(A_i) = \big(|A_j| - c_i(A_j)\big) - \big(|A_i| - c_i(A_i)\big) = w_{\mathcal{A}}(i,j) - |A_i| + |A_j|.$$

Summing along a path $(i_1, \ldots, i_r)$, the terms $|A_{i_t}| - |A_{i_{t+1}}|$ telescope to $|A_{i_1}| - |A_{i_r}|$; along a cycle they telescope to 0. The characterisation of envy-freeability by the absence of positive cycles (Theorem 10) then transfers, and the formula for $\ell_{\mathcal{A}}(u)$ follows from Theorem 11. □

Remark 17 (Why Conjecture 2 is not a corollary of [BKNS22])**.** Theorem 16 says the chore problem is the goods problem of [BKNS22] *plus a cardinality-balancing requirement*. A sufficient condition falls straight out: since $\tilde{d}_{\mathcal{A}}(u, v) \leq \tilde{p}_u - \tilde{p}_v$ for any envy-eliminating $\tilde{p}$, an allocation solving the goods instance in which the quantity $|A_i| + \tilde{p}_i$ has spread at most 1 across agents satisfies $\ell_{\mathcal{A}} \leq 1$. But [BKNS22] offers no control whatever on bundle cardinalities — its output need not even be EF1, by its own Appendix C — so the requirement is not met by invoking it as a black box. The transform also shows the chore problem is at least as expressive as the goods problem, so no easier.

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